Code No: 117FZ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, September - 2021
OPERATIONS RESEARCH
(Common to ME, CSE, MIE)

Time: 3 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) What is a model? Define Iconic model.
b) Find the maximum value of $z = 7x_1 + 10x_2$ by graphical method.
subject to the constraints
$x_1+x_2 \leq 30000$
$x_2 \leq 12000$
$x_1 \geq 6000$
$x_1 \geq x_2$
$x_1, x_2 \geq 0$ [5+10]

2. Solve the following LP - problem by SIMPLEX Method.
Minimize $z = x_1 - 3x_2 + 2x_3$
Subject to
$3x_1 - x_2 + 3x_3 \leq 7$
$-2x_1 + 4x_2 \leq 12$
$-4x_1 + 3x_2 + 5x_3 \leq 10$
x_1 ; x_2 ; and $x_3 \geq 0$ [15]

3. Solve the following transportation problem where all entries are unit costs. [15]

D_1	D_2	D_3	D_4	D_5	D_6	Available
O_1	73	40	9	79	20	8
O_2	62	93	96	8	13	7
O_3	96	65	80	50	65	9
O_4	57	58	29	12	87	3
O_5	56	23	87	18	12	5
Required	6	8	10	4	4	

4. Five jobs 1, 2, 3, 4 and 5 are to be assigned to five machines A, B, C, D and E. The cost of assigning these jobs to the machines in rupees is given in the following matrix. Determine the optimal assignment algorithm so as to minimize the total cost. Calculate the total cost of optimal assignment. [15]

Jobs	Machines				
	A	B	C	D	E
1	6	7	5	9	4
2	7	5	10	9	6
3	5	4	3	6	5
4	8	3	5	6	4
5	4	7	5	6	6

5. There are nine jobs, each of which must go through two machines P and Q in the order PQ, the processing times (in hours) are given below:

Machine	Job(s)								
	A	B	C	D	E	F	G	H	I
P	2	5	4	9	6	8	7	5	4
Q	6	8	7	4	3	9	3	8	11

Find the sequence that minimizes the total elapsed time T. Find T, also calculate the total idle time for the machines in this period. [15]

6. A firm is considering replacement of a machine, whose cost price is Rs, 1,20,000, and the scrap value is Rs. 20,000. The running (maintenance and operating) costs of the machine are as follows:

Year	1	2	3	4	5	6	7	8
Running cost	2,000	5,000	8,000	12,000	18,000	25,000	32,000	40,000

When should the machine be replaced? [15]

7. Determine optimum mixed strategies for each of the players A and B. there is no saddle point. Use dominance principle. [15]

$$\begin{array}{c} \\ A \end{array} \begin{array}{c} \begin{array}{cccc} B & & & \end{array} \\ \begin{bmatrix} 3 & 4 & 8 & 9 \\ 1 & 2 & -2 & 10 \\ 6 & 5 & 12 & 4 \\ 2 & 1 & 0 & 2 \end{bmatrix} \end{array}$$

8. A self-service store employs one cashier at its counter. Nine customers arrive on an average every 5 minutes while the cashier can serve 10 customers in 5 minutes. Assuming Poisson distribution for arrival rate and exponential distribution for service rate, find
a) Average number of customers in the system.
b) Average number of customers in queue or average queue length.
c) Average time a customer spends in the system.
d) Average time a customer waits before being served. [15]

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